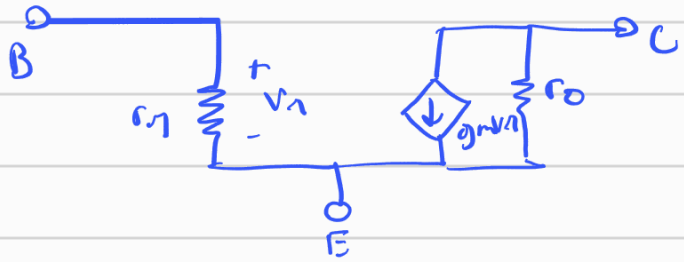
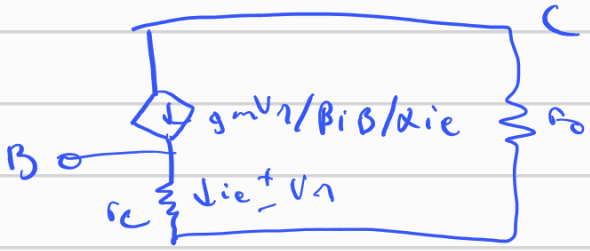
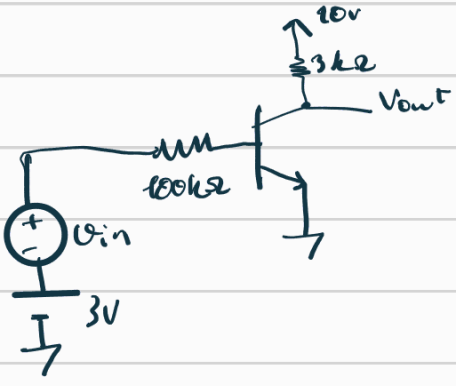
Uygulama



$r_o = \frac{V_A}{I_C}$

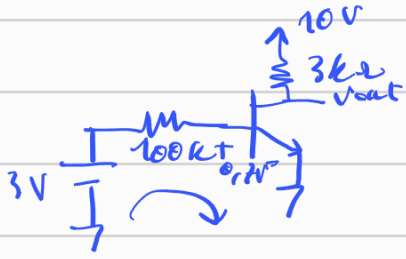




Analyze the circuit at left. Assume $\beta = 100$. $V_A = 100V$. Calculate the voltage gain $\left(\frac{v_{out}}{v_{in}}\right)$, R_{in}, and R_{out}

* Determine the type of amplifier
  - Common-Emitter (use π model)
* DC analysis (to find small signal parameters)

@D



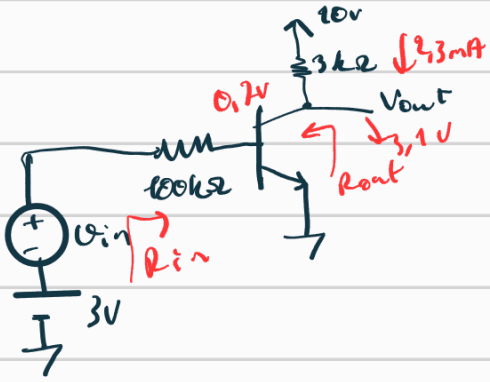
$-3V + 100k I_B + 0,7 = 0$

$100k I_B = 2,3V$

$I_B = 23\mu A$

$I_E = 2,32mA$

$I_C = 2,3mA$



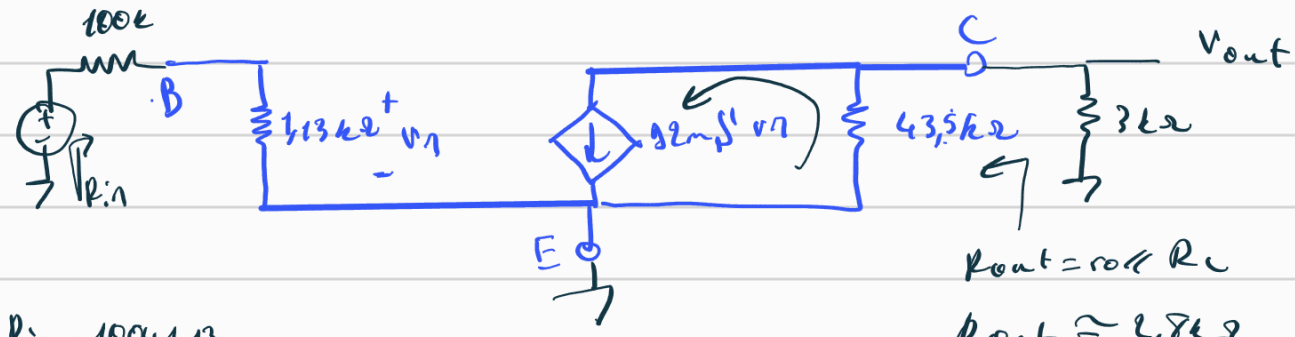
$V_{CB} > 0,4$ (for linear)

$r_o = \frac{100V}{2,3mA} \approx 43,5k\Omega$

$g_m = \frac{I_C}{V_T} = \frac{2,3mA}{25mV} = 92mS$

$r_e = \frac{25mV}{2,32} \approx 11,2\Omega$

$r_\pi = 1,13k\Omega = (\beta+1) r_e$



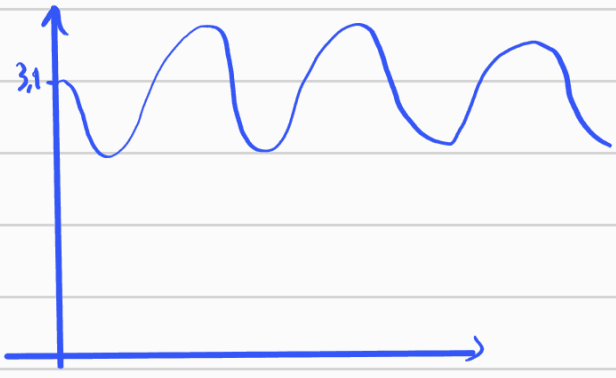
$R_{in} = 100 + 1,13$

$= 101,13 k\Omega$

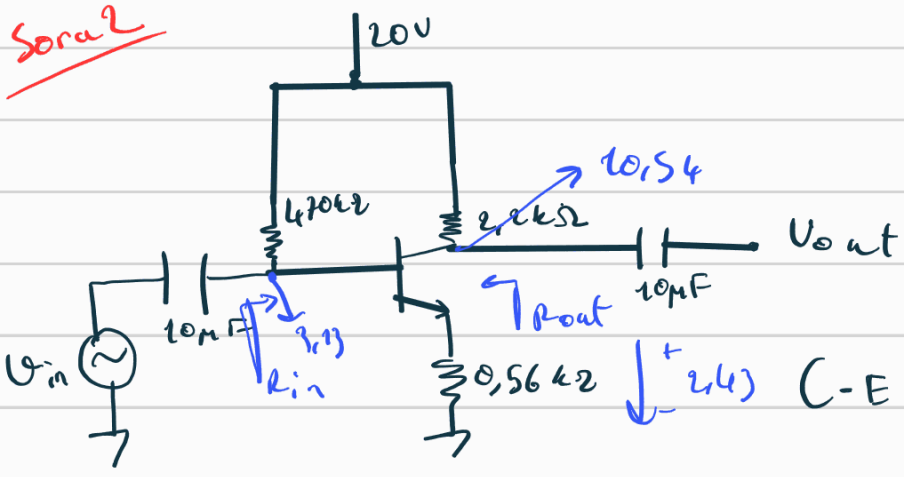
$V_{out} = -2,6 k\Omega \times 92\ S \cdot v_1$

$v_1 = \frac{v_{in} \times 1,13k}{101,3 k\Omega}$

$$V_{out} = \frac{-252,6 \cdot v_{in}\ 1,13k}{101,3k} = -2,68\ V/V$$



Soal 2



C-E with R_E (T model)

$\beta = 120$

$V_A = \infty$

DC

$$-20 + 4{,}20k\,I_B + 0{,}7 + 121 \cdot 0{,}56k\,I_B = 0$$

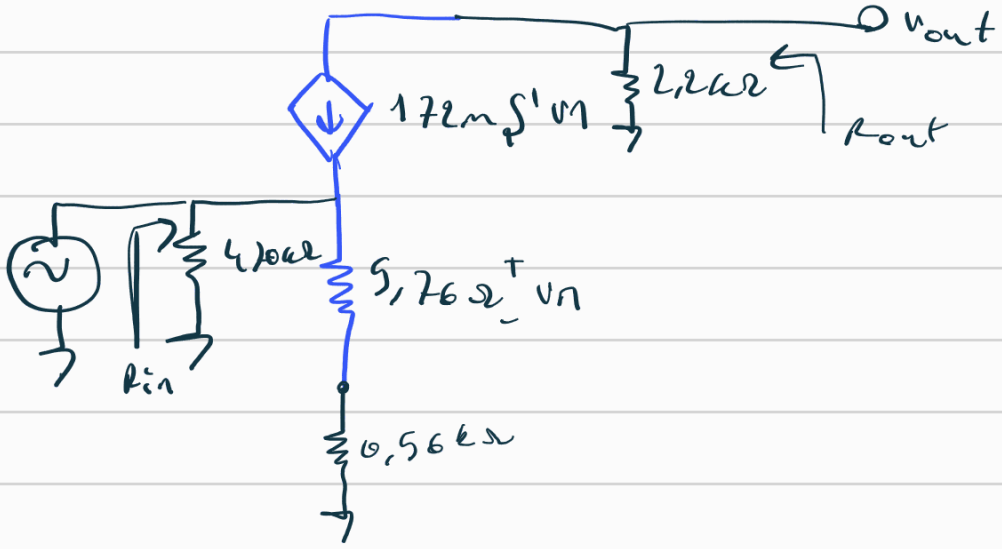
$$I_B = 0{,}035\,mA$$
$$I_C = 4{,}3\,mA$$
$$I_E = 4{,}34\,mA$$

$$V_C = 20 - 2{,}2k \times 4{,}3m\,A = 10{,}54$$

$$r_e = \frac{25\,mV}{4{,}34mA} = 5{,}76\,\Omega$$

$$g_m = \frac{4{,}3}{25} = 172\,mS$$

$$r_o = \infty$$



$$R_{in} = 4{,}20k\,\|\,\underbrace{(121)(569{,}76\,\Omega)}_{68{,}4k\Omega}$$

$$R_{out} = 2{,}2k\Omega$$

$$R_{in} = 59{,}8\,k\Omega$$

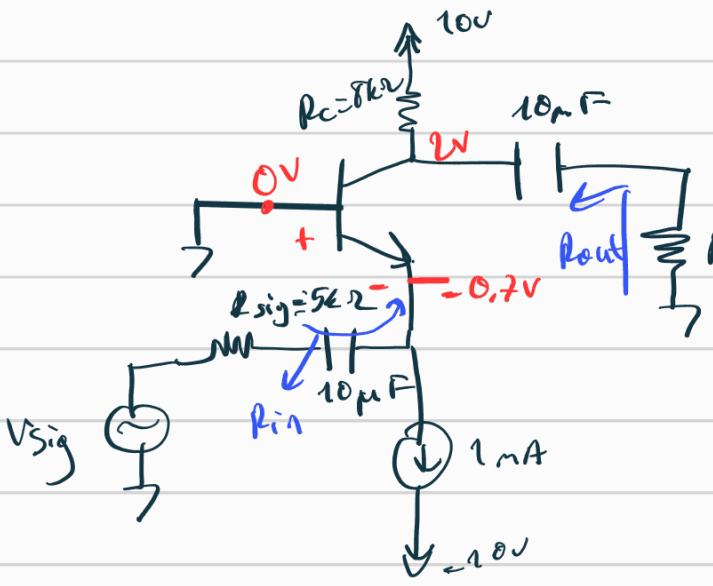
$$v_{out} = -172mS \cdot 2{,}2k \cdot v_\pi$$

$$v_\pi = \frac{v_{in}}{565} \times 5{,}76$$

$$v_{out} = -378{,}4\,v_{in} \times 0{,}01$$

$$\frac{v_{out}}{v_{in}} \cong -3{,}784\,[V/V]$$

Sorry



R_{in}, R_{out}, A_v $\left(\frac{v_{out}}{v_{sig}}\right)$

CB Amplifier

$\beta = 100$ $V_A = \infty$

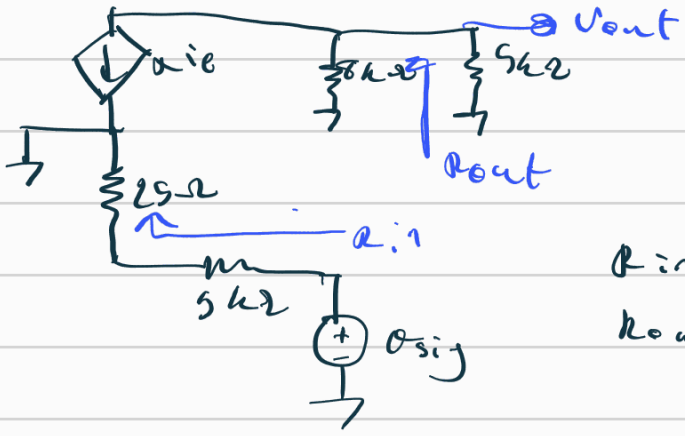
$I_C = 1mA$

$v_t = 25mV$

$r_e = 25\Omega$

$r_\pi \cong 2,5k$

$g_m = \frac{1mA}{25mV} = 40mS$



$R_{in} = r_e = 25\Omega$

$R_{out} = 5k\Omega$

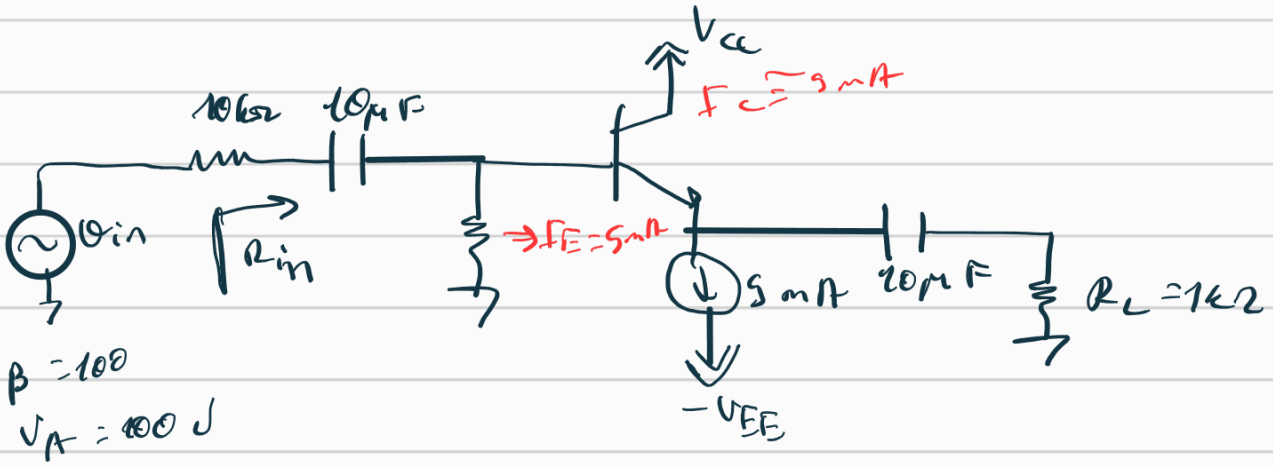
Common emitter de sadece eğaz bark. var.

$v_{out} = -3,07k\Omega \cdot i_e$

$i_e = \frac{-v_{sig}}{5,025k}$

$\frac{v_{out}}{v_{sig}} = \frac{3,07 \cdot 0,99}{5,025} = 0,6 \left[V/V\right]$

Sorn

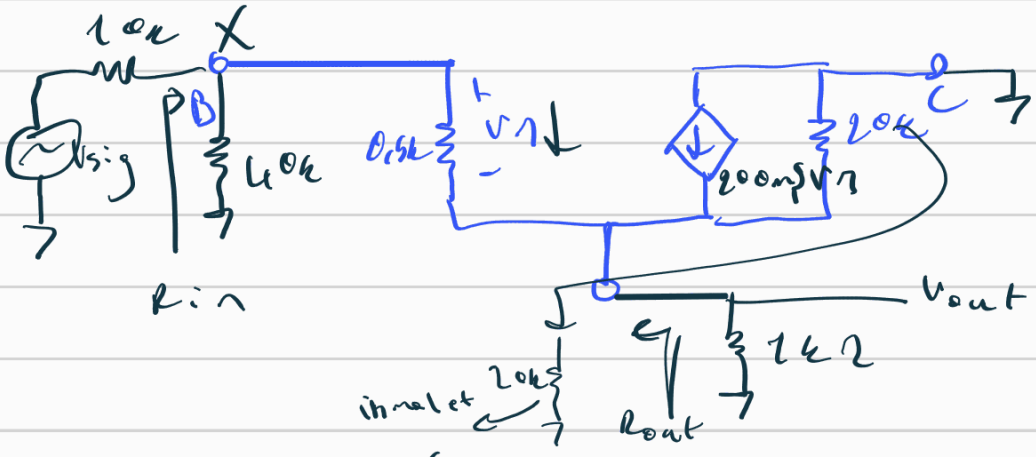


$$r_e = \frac{25\,mV}{5\,mA} = 5\,\Omega$$

$$r_\pi = 0{,}5k$$

$$g_m = \frac{5\,mA}{25\,mV} = 200\,mS$$

$$r_o = \frac{100\,V}{5\,mA} = 20k\Omega$$



$$R_{in} = 40k \,//\, (0{,}5k + 10\,k)$$

$$R_{in} \simeq 28{,}5\,k\Omega$$

$$R_{out} = \frac{(10k \,//\, 40k) + 0{,}5k}{\beta + 1}$$

$$R_{out} \simeq 80\,\Omega$$

$$V_{out} = \left(\frac{V_\pi}{0{,}5k} + 200\,mS\,V_\pi\right) 1k$$

$$= 2V_\pi + 200V_\pi$$

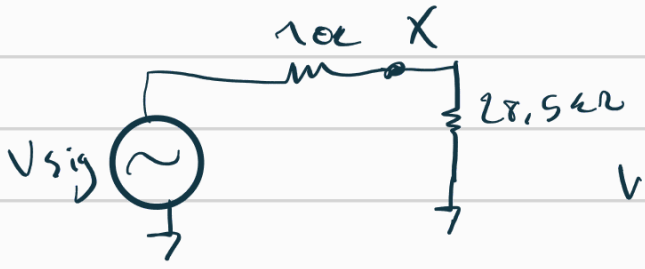
$$= 202\,V_\pi$$

$R_{sig}=0$ is c

$\Rightarrow V_{sig} = V_{x1} + V_{out}$

$$\frac{V_{out}}{V_{sig}} = \frac{202 V_1}{203 V_1} \approx 0,99 \left[\frac{V}{V}\right]$$

$R_{sig}=10 k\Omega$



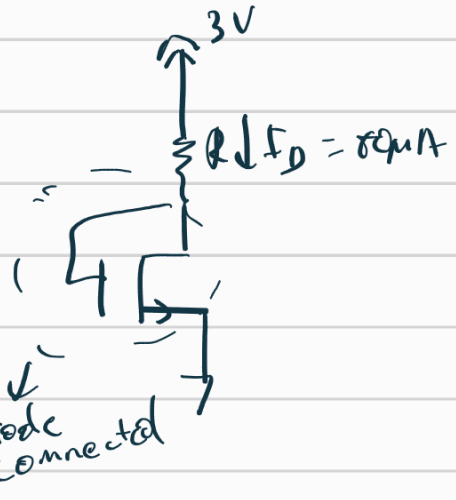
$$V_X = \frac{28,5}{38,5} V_{sig} = 0,74\, V_{sig}$$

$$V_1 = V_X - V_{out}$$
$$V_1 = 0,74\, V_{sig} - 202 V_1$$
$$203 V_1 = 0,74\, V_{sig}$$
$$V_{sig} = 273,4 V_1$$

$$\Rightarrow \frac{V_{out}}{V_{sig}} = \frac{202 V_1}{274,3 V_1}$$
$$\approx 0,734 \left[\frac{V}{V}\right]$$

Design the circuit given below to obtain a current of 80 μA.

Find the value of R_D and V_D. Let the NMOS have $V_T = 0,6$ V, $\mu_n C_{ox} = 200 \mu A/V^2$, $L = 0,8 \mu m$, $W = 4 \mu m$. Neglect channel length modulation.



$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$$

$$80 mA = \frac{1}{2} 200 \frac{\mu A}{V^2} \frac{4}{0,8} (V_{ov})^2$$

$$\frac{160}{1000} (V_{ov})^2 = 0,16$$

$V_{ov} = 0,4$ ~~$= -0,4$~~

Sat

$V_{GS} - V_T = 0,4V$

$V_{GS} = 0,4V + 0,6V = 1V$

$V_G = 1V + V_S = 1V$

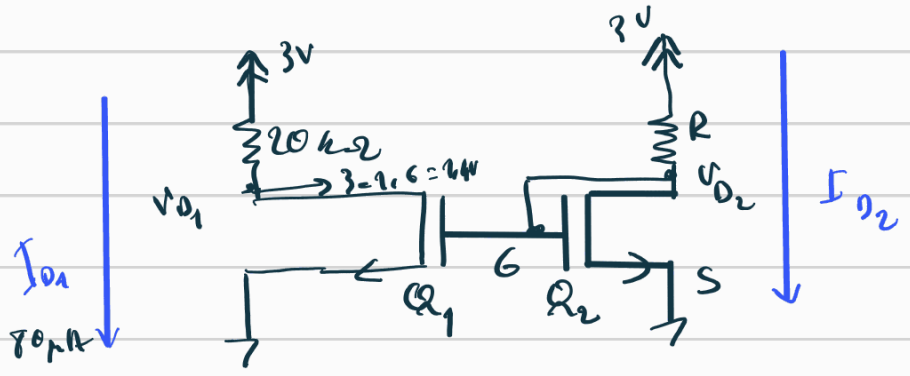
$V_G = V_D = 1V$

$V_{GS} \geq V_T$

$V_{DS} \geq V_{GS} - V_T$

$R_D = \frac{3-1}{80 \cdot 10^{-6}}$

$= \frac{2}{80 \cdot 10^{-6}} = \frac{2 \cdot 10^6}{80} = 25k\Omega$



$I_{D1} = 80\mu A$

$V_{D1} = ?$

$I_{D2} = ?$

Q_1 identical to Q_2

$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS1} - V_t)^2$

$I_{D2} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS2} - V_t)^2$

$I_{D1} = I_{D2} = 80\mu A$

$R_{min} \leq R \leq R_{max}$

$V_T = 0,6V$

Sat

$V_{GS} \geq V_T$

$V_{DS} \geq V_{GS} - V_T$

$V_D = V_G - V_T$

$V_G = V_D + V = 2V$

Sat

$\frac{3-0,6}{80\mu A} = 30k\Omega$

$\frac{3-2}{80\mu A} = 12,5k\Omega$

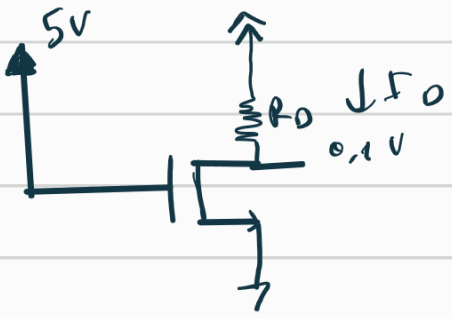
$V_{DS} \leq V_{GS} - V_T$ } triode

$V_{GS} \geq V_T$ } SAT

$V_{DS} \geq V_{GS} - V_T$ (V_{OV})

$V_{GS} < V_T$ } cutoff

Design the circuit to obtain a drain voltage of 0,1 V
What is the effective resistance between drain and source at this point $V_T = 1V$ and $\frac{1}{2} \mu_n C_{ox} \frac{W}{L} = 1 mA/V^2$.



$V_{DS} < V_{GS} - V_T$ $\quad 0,1 < 5V$ } transistor
$\times$

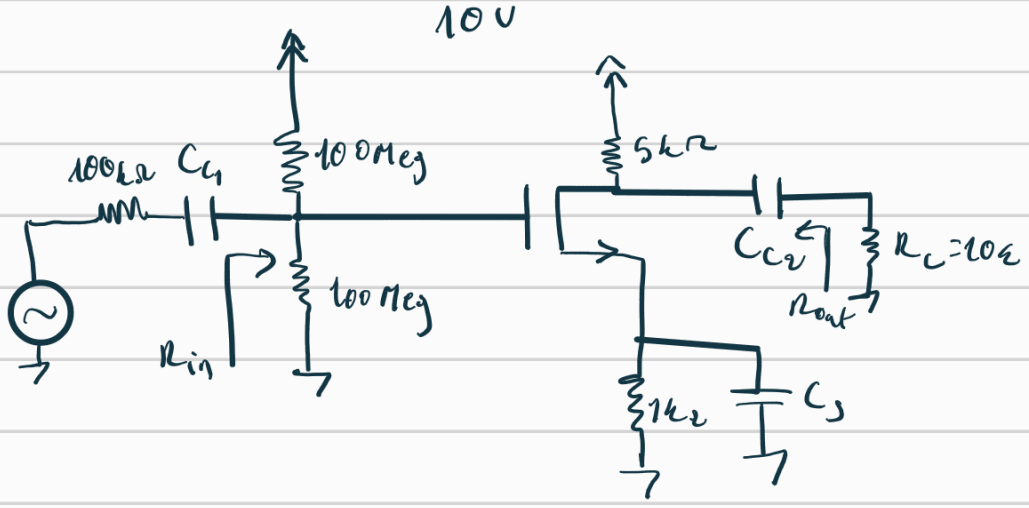
$I_D = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T - \frac{V_{DS}}{2}) V_{DS}$

$\frac{V_{DS}}{I_D} = r_{DS} = \frac{1}{K} =$

$\rightarrow \frac{1000}{79} \simeq 125 \Omega$

R_D 'ideal' = ?

Soln



$|V_T| = 2V$, $\mu_n C_{ox} = 125 \mu A/V^2$, $W = 4\mu m$, $L = 1\mu m$

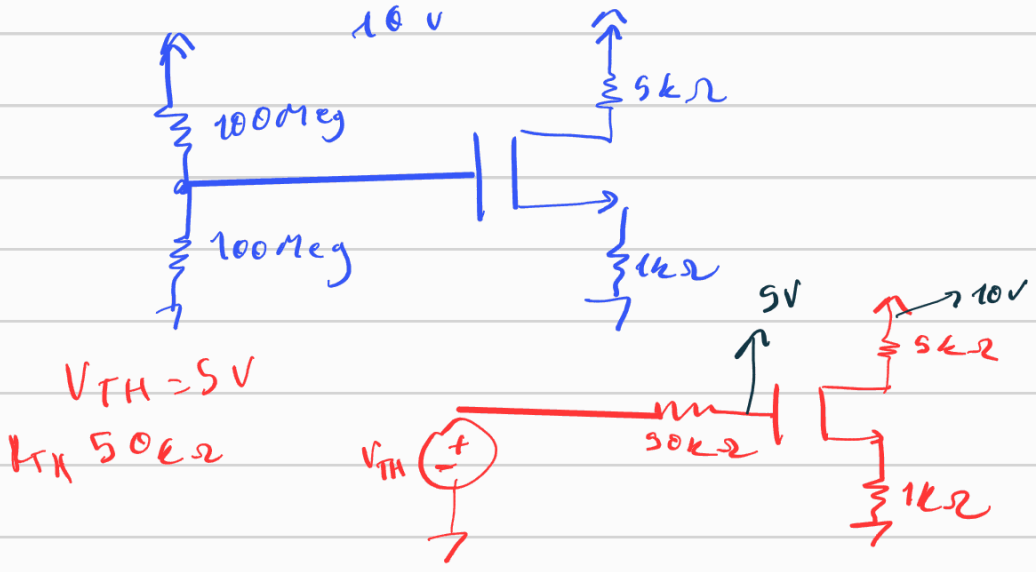
- Perform DC analysis, calculate small signal parameters
- Estimate the voltage gain $\left[\frac{V_{out}}{V_{sig}}\right]$, R_{in}, and R_{out}

$V_A = 50\ V$

DC: $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$

Small signal: $r_o = V_A / I_D$

$g_m = 2 I_D / V_{OV}$



$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$

$V_G = 5V$, $V_S = I_D$

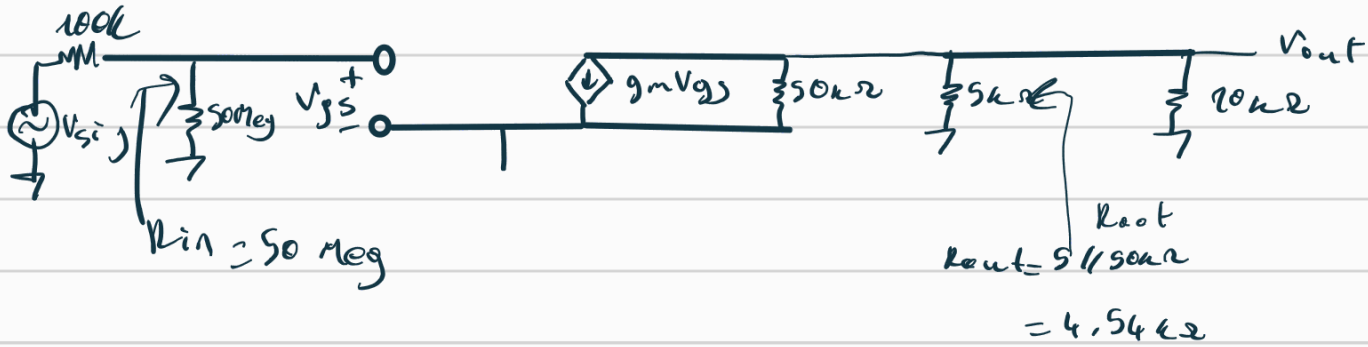
$I_D = \frac{1}{2} 125 \mu A/V^2 \cdot 4 (5 - I_D - 2)^2$

$I_D = 250 \mu A/V^2 (5 - I_D - 2)^2$

$I_D = 1mA$

$r_o = 50/1mA = 50k$

$g_m = \frac{2 \cdot 1\ mA}{2}$ $\quad g_m = 1mS$

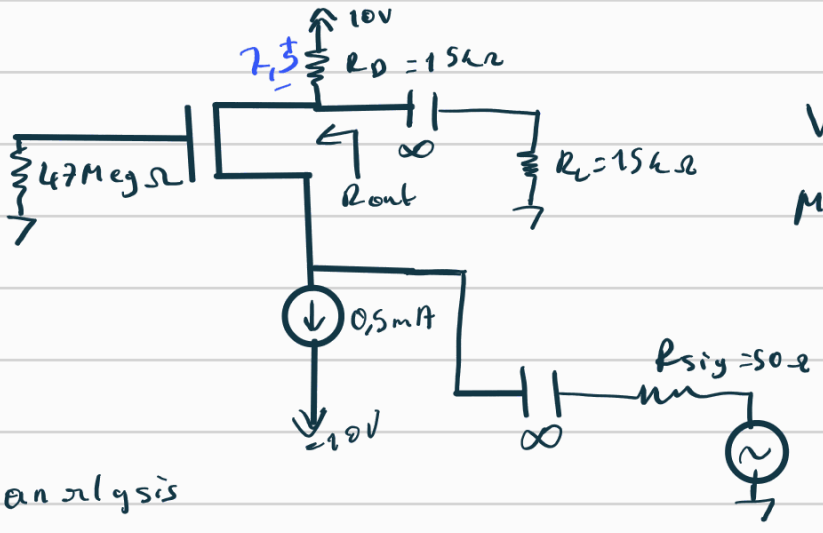


$$v_{gs} = \frac{v_{sig} \cdot 50meg}{50,1} \simeq v_{sig}$$

$$v_{out} = -(\underbrace{50k \,//\, 5k \,//\, 10k}_{= 3,12k}) \cdot 1mS \cdot v_{sig}$$

$$V_{out}/V_{sig} = -3,12$$

Sorusu



$V_t = 1,5V$

$\mu_n C_{ox} \frac{W}{L} = 1 mA/V^2$

$|V_A| = \infty$

a) DC analysis

b) Draw small signal equivalent circuit

c) Calculate gain, Rin, Rout

$I_D = 0,5mA$

Common-Gate (T mode)

$V_D = 2,5V$

$V_G = 0V$

$0,5mA = 1/2 \; 1 mA/V^2 (V_{ov})^2$

$V_{ov} = 1V$

$V_{GS} - V_T = 1V$

$0 - V_S - 1,5 = 1V \qquad V_S = -2,5V$

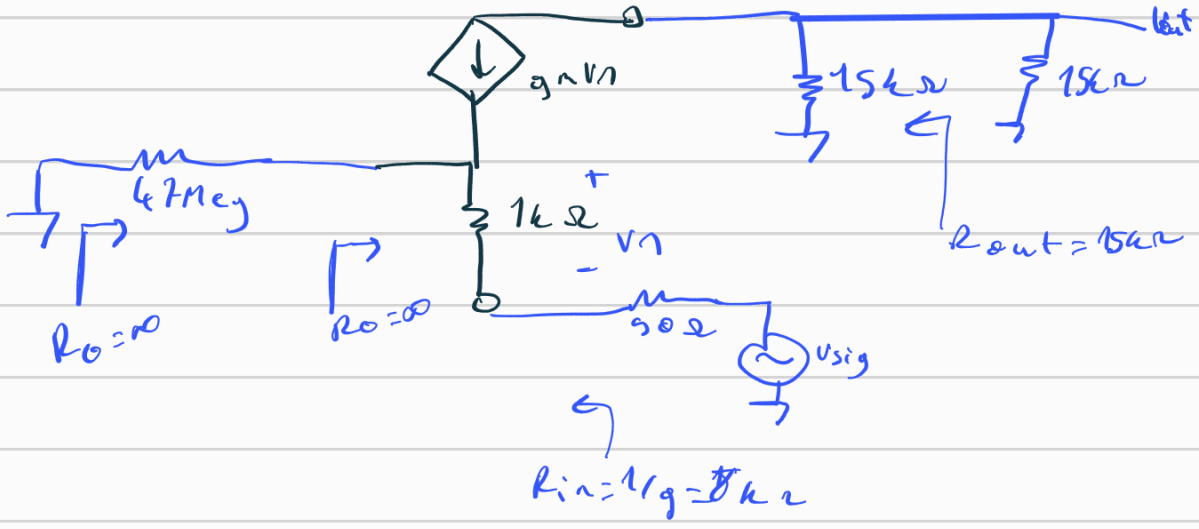
$V_{DS} \geq V_{ov}$

$5V \geq 1V$ sat

$r_o = \infty$

$g_m = \frac{2 I_D}{V_{ov}} = \frac{2 \cdot 0{,}5\,mA}{1\,V}$

$g_m = 1\,mS$



$V_{out} = -7{,}5k \cdot 1mS \, V_n$

$= -7{,}5\, V_n$

$V_{sig} \approx V_n$

$\frac{V_{out}}{V_{sig}} = -7{,}5 \left[V/V \right]$